MARKETING OPERATIONS · CREATIVE COMPLIANCE · SEPTEMBER 2026

Multi-Channel Creative Proofing

The expert primer. Online proofing SaaS, automated artwork & text comparison, customer-communications validation, channel-specific QA, financial-services marketing compliance, and the AI layer being bolted onto all of it - what exists, what it actually does, what it costs, and what nobody has built yet.

Prepared for Brandon

Built from three adversarially-verified research passes (321 sub-agent calls, 80–135 claims extracted per pass, ~60% confirmed after 3-vote adversarial verification) plus four direct-research deep dives on operational vocabulary, workflow reality, and documented enforcement costs.

Every claim below is dated and sourced. Vendor marketing copy is labelled as such. Unresolved questions are stated as unresolved - not guessed at.

Contents

00 How to use this document

The confidence key - read this before you quote anything

Every substantive claim in this document carries one of four tags. This report was built by adversarially verifying claims across three research passes - independent reviewers tried to refute each one before it was allowed to stand - but even confirmed claims vary in what kind of thing they are:

TAG	WHAT IT MEANS
VERIFIED	Confirmed against a primary source - a regulator's own text, an SEC filing, raw vendor HTML fetched and parsed directly, or a government enforcement order. Survived a 3-vote (or better) adversarial check.
VENDOR CLAIM	What a company says about its own product. Presented because it establishes what a vendor is *positioning*, not because the number is true. ROI and efficiency percentages in this category failed verification almost universally - see §12.
GAP	We looked and did not find a reliable answer. Stated as a gap, not papered over with an estimate.
INFERENCE	Our synthesis or translation into your vocabulary, clearly separated from what a source actually said.

Three passes, if you want the Kwik-style structure

Pass 1 (10 min) - skim §01, §03, and the nine-row table in §13. That gives you the shape of the market and exactly which of your use cases already have a vendor answer.

Pass 2 (25 min) - read §04–09 in full. This is the vendor landscape, segment by segment, with pricing and what the AI layer actually does versus claims to do.

Pass 3 (20 min) - read §10–12. This is the "why" layer: how review actually works inside an enterprise, what it costs when it fails, and what AI genuinely changes versus what is marketing.

YOUR DOMINANT QUESTION

"Which of my nine use cases already has a vendor, and which am I the first to build?"

Section 13 answers this directly. The short version: rules-checklist validation and cross-format ingest are commodity. Journey validation - click a code, render the destination, compare against the approved version - is closed by nobody.

01 The whole field in 60 seconds

The things that are actually true in September 2026

1. "**Marketing proofing**" **is not one market.** It splits into at least six segments with different buyers, price points, and technology: workflow proofing SaaS, automated artwork/text comparison, customer-communications management, channel-specific QA tooling, the marketing-ops system of record, and regulated-industry compliance review. They share a customer base but almost nothing else.
2. **Rules-based AI validation moved down-market in the last twelve months.** Checking an asset against a compliance rulebook used to require a specialist vendor. It now ships as metered credits inside $199–$329/month proofing tools - Filestage's Review Agents, Ziflow's ReviewAI, PageProof Intelligence, GoVisually's Compliance AI Suite all shipped or matured in 2025–2026.
3. **The category consolidated hard in the same window.** Veralto bought GlobalVision for ~$195M (April 2026, confirmed by SEC 10-Q). Validity bought Litmus (April 2025). Sinch is retiring Email on Acid into Mailgun Inspect. The pattern is incumbents absorbing point solutions, not new entrants disrupting incumbents.
4. **Every AI product that ships stops short of autonomous sign-off.** Ziflow outputs "AI-suggested" comments. GlobalVision outputs a risk score. Filestage's agents output suggestions. All of them feed a human review stage. Schlafender Hase states this as a design philosophy outright: "*TVT shows you all differences. The expert determines what is right or wrong.*" Agentic *detection* is real. Agentic *sign-off* is not being sold by anyone, and probably shouldn't be - see §12.
5. **Every quantified vendor ROI claim we tested failed verification.** Not "was unverifiable" - actively failed a 3-vote adversarial check. There is no independent benchmark anywhere in this category for AI review accuracy, false-positive rate, or time saved. A buyer today cannot compare vendors on evidence.
6. **Your hardest use case - click a QR/URL, render the destination, screenshot it, compare against the legally approved version - is unclosed by every vendor examined.** The rendering and pixel-diff primitives exist (Ziflow does both, separately). Nobody has wired them into an agent that traverses a code and validates the destination.
7. **The regulatory record is the best evidence for why this category exists at all.** We found 40+ cited enforcement actions - FTC, FINRA, SEC, FCA, ASA - where the violation *was* the marketing creative or a disclosure inside it. Several turn on font size and layout, not wording. See §11.

The one-sentence version

Workflow proofing is a commodity that AI is being bolted onto as the monetization layer; deep content comparison remains a specialist, opaque-pricing business now owned by industrial conglomerates; and the hardest, most valuable jobs - cross-format variant validation and destination journey checks - have no vendor answer yet.

02 The core mental model

Every proofing system on earth, regardless of segment, answers the same five questions in the same order. Use this to evaluate any vendor pitch you hear, including ones not in this report.

SOURCE - what is truth?

Before anything can be validated, someone has to define the reference. In your world this is the **copy deck** (the approved words) and the **annotated PDF from legal** (the approved words *plus* the sign-off marks). In pharma it's the current FDA-approved prescribing information. In Messagepoint's Medicare product it's the CMS model document. Whoever owns SOURCE owns the standard everything else is measured against - and if SOURCE itself is out of date, every downstream check is validating against the wrong thing. This is Segment C's entire insight: Messagepoint splits SOURCE into **Black Text** (must appear verbatim) and **Blue Text** (must appear only if this variant's attributes require it) - the cleanest available decomposition of "what does 'correct' even mean for a produced variant."

RENDER - what did we actually build?

The produced asset, in its final channel-specific form: the PDF, the sent email, the live web page, the printed piece, the authenticated post-login screen. This is where **channel coding** happens - the produced asset carries tracking/offer/version identifiers that must themselves be validated, not just the creative content. RENDER is also where format breadth lives, and it is the one part of this whole problem that is genuinely solved: Ziflow ingests 1,200+ file types on its free tier.

COMPARE - how do we detect divergence?

Three technically distinct approaches exist in this category and vendors rarely say which one they use:

- **Deterministic pixel/character comparison** - GlobalVision's pixel-by-pixel overlay and character-by-character text diff; the open-source visual-regression tier (Argos, Chromatic, BackstopJS) that explicitly rejects ML for auditability.
- **Text/content-level comparison across formats** - Schlafender Hase's TVT, which compares Word-to-PDF or XML-to-Excel at the content level regardless of layout.
- **AI/LLM-based rule interpretation** - GlobalVision's CheckAI, Ziflow's ReviewAI, Adobe GenStudio's Content Check - applied on top of, not instead of, the deterministic layer.

JUDGE - which divergences matter?

Not every detected difference is a defect. A font-rendering artifact is noise; a missing disclaimer is a violation. This is where false positives live, and where the checklist/rulebook concept (your use case: "*validate creative against a rules checklist*") does its work - Filestage's named agents (Forbidden Terms, Mandatory Image, Compliance-vs-uploaded-context) are all JUDGE-layer products sitting on top of a comparison engine.

PROVE - how do we show a regulator we checked?

The audit trail: who approved what, when, against which version. Section 10 shows this is not a convenience feature - it is directly mandated by 21 CFR 11.50 (pharma) and FINRA Rule 2210(b)(4) (financial services), both of which require the *meaning* of a signature (review, approval, authorship) to be embedded in the human-readable record itself, not just logged in a database somewhere.

> **HOW TO USE THIS MODEL IN THE WILD**

> When a vendor says "AI-powered compliance review," ask which of these five layers it actually touches. Almost everything on the market today only does JUDGE - it takes someone else's comparison output and applies rules to it. Very few vendors do COMPARE well across heterogeneous formats (Schlafender Hase and GlobalVision are the exceptions, and neither publishes prices). Nobody found in this research does PROVE well across formats and channels in one system.

03 The landscape map

Six segments, sharing customers but little else. The boundaries blur exactly where the AI layer is being added - a proofing SaaS tool with a compliance agent starts to look like a compliance vendor; a compliance vendor with workflow routing starts to look like proofing SaaS.

SEGMENT	WHAT IT IS	WHO BUYS IT	PRICE RANGE
A - Online proofing	Route assets for markup, annotation, approval	Any marketing/creative team	$2.4K–$5K/yr entry, Enterprise custom
B - Artwork/text comparison	Deterministic compare of built asset vs. source	Packaging, pharma, print	Opaque - no published pricing
C - Customer comms (CCM)	Compose & validate variable/personalized documents	Insurance, healthcare, banking ops	Opaque - enterprise sales-led
D - Channel-specific QA	Email rendering, link checking, visual regression, a11y	Email/web/marketing ops teams	$0–$199+/mo, often metered
E - System of record	AEM, Aprimo, Workfront - where assets live	Enterprise IT + marketing ops	N/A - the platform, not the check
F - Regulated compliance	FINRA/SEC marketing-rule pre-publication review	Broker-dealers, RIAs, insurers	Opaque - enterprise sales-led

WHERE THIS REPORT STOPS SHORT

Segment E (§08) and the AI brand-governance corner of Segment F (Writer.com, Grammarly, Acrolinx, Frontify) did not yield verified data despite three research passes. Market size for the combined category also did not surface from any credible analyst (Gartner/Forrester/IDC) in three attempts - we are treating that as a real gap in public data, not a search failure. See §17 for the full ledger.

04 Segment A - Online proofing & creative review

The commodity core of the category: route an asset, collect markup, version it, approve it. Base functionality - proofs, reviewers, storage, versions - is sold as unlimited across every vendor examined. AI is universally the paid upsell layered on top.

VENDOR	AI FEATURE (2025–26)	GATING	ENTRY PRICING
Ziflow	ReviewAI - automated pass/fail against checklist criteria	Enterprise-only, early access, paid add-on	Free (2 users) / $199 & $329/mo annual (15–20 users) / Enterprise custom
Filestage	Review Agents: Text, Grammar, Image, Forbidden Terms, Compliance-vs-context-docs, Barcode/QR	50 non-rollover credits/mo, Business tier+	Free / $199 & $329/mo
PageProof	PageProof Intelligence: Smart Check, Smart Compare, PI Content Credentials Check	Metered "PI credits": 250/500/1,000 by tier	$2,499 & $4,999/yr flat, unlimited reviewers
GoVisually	"CPG Compliance AI Suite" - label compliance	Flat premium SKU, not bundled	$16–$33/user/mo base; AI suite $499/mo separate

What "1,200+ file types" actually means

Ziflow's free tier VERIFIED ingests images, Adobe CC native files, Office documents, video/audio, static and live websites at any resolution, and HTML5 banners - with pixel-level comparison on static and video proofs included even for free users. This genuinely answers your first use case (cross-channel format breadth) at the *ingest* layer. But two caveats matter: it's "1,200+ file types" as a round, unauditable vendor figure - treat it as "advertises," not "supports" - and multi-asset proofs (reviewing several files as one campaign) are tier-gated, so free and Standard users get one asset per proof. INFERENCE Ingesting formats is not the same as comparison intelligence across them - format breadth is table stakes; cross-format validation logic is not.

The 10-page ceiling

VERIFIED Ziflow's ReviewAI "currently works on static assets (images, PDFs) up to 10 pages," per its own help center. It does not touch video, motion, web, or long-form content - precisely where AI-generated marketing volume is growing fastest. Its Brand Standards module (logo, tagline, font, color enforcement) and Change Verification module are both marked "coming soon" as of this research.

Market size

GAP No credible analyst estimate (Gartner, Forrester, IDC) for the online-proofing / creative-operations category surfaced across three research passes. Any figure you encounter citing "the creative operations market is worth $X billion" almost certainly traces to a research-mill publisher (Grand View, MarketsandMarkets, and similar) whose methodology could not be independently verified - treat those numbers as decorative, not load-bearing.

05 Segment B - Automated artwork & text comparison

The oldest and most rigorous part of this landscape. GlobalVision and Schlafender Hase have been doing "compare the built asset against the approved source, character by character and pixel by pixel" for twenty-plus years, mostly for pharmaceutical packaging and print. This is the closest existing analogue to your use case #4 - and the one segment where a real acquisition price gives us a genuine market-size anchor.

> **THE VERALTO ACQUISITION - THE STRONGEST EVIDENCE IN THIS ENTIRE REPORT**
>
> VERIFIED, SEC 10-Q On April 7, 2026, Veralto Corp (NYSE: VLTO, the 2023 Danaher spinout) completed the acquisition of 100% of GlobalVision for approximately **CAD $270 million (~USD $195 million)**, net of cash acquired, per its Form 10-Q for the quarter ended July 3, 2026. GlobalVision is now integrated into Veralto's Product Quality & Innovation segment and brand-endorsed as "GlobalVision, an Esko Company." ~$134M of goodwill was preliminarily recorded. Veralto's own filing language is precise and worth quoting directly: GlobalVision "leverages its core proprietary deterministic technology with AI-augmented functionality."

How the technology actually works

GlobalVision's product family VERIFIED splits cleanly: **Verify Compare** does character-by-character text comparison "no matter the layout" plus pixel-by-pixel graphics overlay, barcode grading to ISO/ANSI/CEN standards, and braille verification against the Marburg Medium Font Standard. **GVD** pairs the same engine with flatbed/roll scanners for off-press print inspection. **Verify API** is REST/stateless and integrates natively with Hybrid CLOUDFLOW and Esko Automation Engine - both packaging-prepress systems, notably *not* AEM, Aprimo, Workfront, or Jira. **Verify CheckAI**, GA since March 2026, is the new AI layer: it inspects packaging/labeling against custom rules and named regulatory authorities (FDA, EU), auto-detects product type and market, and outputs a compliance score. It is explicitly positioned as "compliance interpretation rather than visual asset verification" - complementary to, not a replacement for, the deterministic engine.

Schlafender Hase's TVT VERIFIED is genuinely the closest match to "validate against a source copy deck" among everything researched: it compares documents *across different file formats and layouts* - Word-to-PDF, XML-to-Excel, many-to-one - at the text/content level, character-by-character on Unicode values. A published customer proof point: "cut down their text verification time by over 50% when comparing Word to PDF and Word to Word documents" - which is structurally identical to comparing a copy deck against a rendered PDF. Pixel-level artwork comparison is a separate add-on (TVT Artwork, four sensitivity levels), sold alongside TVT Spelling and TVT Barcode. **Schlafender Hase publishes no prices anywhere.**

> **THE DESIGN STANCE THAT ANSWERS "IS AI ACTUALLY TRUSTWORTHY HERE"**
>
> Schlafender Hase's own published FAQ asks itself the obvious question and answers it in public: "*Does TVT decide what is a mistake during document comparison? No. TVT shows you all differences between your documents. The expert using TVT determines what is right or wrong.*" This is the mature, twenty-year-old answer to the AI trust problem, and every modern AI entrant in this category - without exception - has converged on the same design: detect everything, let a human adjudicate. See §12.

06 Segment C - Customer comms & variable-data proofing

This is where "validate dynamic variables against a variable-data matrix" and "proof every variant" already exist as named, partially-solved problems - just not in marketing-creative vocabulary. Direct mail, statements, and post-login documents are this segment's home turf.

> **MESSAGEPOINT'S MARCIE ASSURE - THE SINGLE BEST COMMERCIAL ANALOGUE TO YOUR CORE JOB**
>
> **VERIFIED (vendor pages + press release, no independent customer data)** Launched June–July 2026, MARCIE Assure validates Medicare Advantage ANOC/EOC documents against CMS model documents on two axes:
>
> - **Black Text** - mandatory CMS model language that must be present and match the latest CMS model documents *verbatim*.
> - **Blue Text** - conditional CMS model language that must be correctly included *based on each plan's structure and benefits* - present if applicable, absent if not.
>
> It separately validates plan details (IDs, names, cost shares, deductibles, authorization requirements) against the Plan Benefit Package (PBP) and plan administrative data files as "sources of truth" - functionally identical to validating against a variable-data matrix in Excel.

INFERENCE The Black Text / Blue Text split is the most useful teaching device in this entire research effort. It decomposes "is this variant correct" into exactly two questions: *must this match exactly*, and *must this be present at all, given what we know about this specific variant*. That's the rule taxonomy any general-purpose validator needs for your use cases 4, 5, and 7 - annotated-PDF-vs-copy-deck matching, multi-version validation, and dynamic-variable checking are all the same underlying problem MARCIE Assure solves for one document type.

The limits that matter: MARCIE Assure is domain-locked to Medicare Advantage ANOC/EOC documents inside Messagepoint's own CCM repository - it is not a general multi-channel engine, and it cannot be pointed at an email, a web page, or a PDF outside that pipeline. No customer count, pricing, or independent review was found anywhere. Neither vendor page discloses the underlying comparison technique (text extraction, OCR, or LLM semantic match) - that detail is simply not public.

GAP The wider CCM category - Quadient Inspire, Smart Communications, OpenText Exstream, Precisely EngageOne, Elixir, Sefas - did not yield verified data on their own variable-data proofing capabilities in this research. Messagepoint is the one data point, not a survey of the category.

07 Segment D - Channel-specific QA tooling

Purpose-built checkers for one channel at a time: email rendering, link integrity, pixel-level visual regression, and accessibility. This is the segment your use cases lean on hardest - and where the biggest unclosed gap in the whole report lives.

Email QA - a consolidating category

VERIFIED Litmus was acquired by Validity (April 10, 2025) and absorbed into "Validity Engage" (announced Feb 11, 2026) - it is no longer a standalone product. Email on Acid (Sinch) is being retired into **Mailgun Inspect**, with mandatory customer migration beginning June 2026. Mailgun Inspect prices at $99/mo (1,000 previews) and $199/mo (2,000 previews), both with unlimited users and API access, plus $0.15/preview overage. Parcel prices per seat ($0/$29/$56/mo) but meters real-inbox rendering separately at $49 for 500 previews, falling to 8–9¢/preview at volume. Bundled into every email QA tool: accessibility checking, HTML problem checking, spellcheck, dark-mode preview - free tier in Parcel's case. Link and image validation are Pro-tier-and-above; spam scoring (SpamAssassin) is typically the top-tier feature.

Link & URL validation

VERIFIED Screaming Frog SEO Spider is the workhorse: a per-user *annual term licence* at $279/licence/year (dropping to $235 at 20+ licences) - explicitly not perpetual; licences expire and revert to a free tier hard-capped at 500 URLs per crawl. Siteimprove, the accessibility/QA incumbent, is privately held by Nordic Capital, which took a majority position in 2020 and, on current board-composition evidence, still holds it into 2026.

Visual regression & pixel-diff testing - the mechanism, precisely

VERIFIED The entire open-source/CI visual-regression tier (Argos, Chromatic, BackstopJS) uses **deterministic raster comparison**, not ML. Argos states this as an explicit design stance, in its own documentation: "*Some tools use AI or ML models to decide whether a change is acceptable. Argos intentionally does not: AI compensates for flakiness, while Argos removes it. Deterministic / Explainable... Review-friendly... Auditable: Approvals have a clear meaning.*" It runs on the open-source `odiff` library. Chromatic compares color distance at the same (x,y) coordinate in YIQ color space. BackstopJS delegates to a Resemble.js fork.

False positives are managed by four documented, non-ML mechanisms: anti-aliasing detection and suppression (on by default in pixelmatch and Chromatic); a tunable mismatch-percentage threshold; proactive pausing of CSS animations, video, and GIFs before capture (Chromatic explicitly cannot pause JavaScript-driven animation - a named residual gap); and hiding dynamic DOM regions before comparison. Argos's baseline-selection logic is genuinely sophisticated: it walks git commit ancestry to find the most recent complete, passed, approved build on an ancestor commit of the current merge base - solving the hardest operational problem in visual regression (what do we compare against?) with an automated, auditable rule rather than a person clicking "set new baseline."

GAP Nobody found in this research applies this software-QA-grade visual regression tooling specifically to marketing creative rather than software UI. That is a genuine, unclaimed white space - see §14.

Accessibility

VERIFIED Deque's axe-core is MPL-2.0 licensed, actively maintained, and embeddable in proprietary products - it is the engine behind Google Lighthouse's accessibility score. Siteimprove notably does *not* use axe; it built its own independent open-source engine, Alfa, on the W3C ACT Rules Format. Do not assume every accessibility vendor is an axe wrapper.

Adobe GenStudio for Performance Marketing - the most significant recent entrant

VERIFIED GenStudio ships a native "Content check" panel that validates generated content simultaneously against (1) customer brand guidelines, (2) per-channel platform guidelines (Meta and LinkedIn named explicitly), and (3) accessibility standards - returning pass/fail per item and a compliance score defined as guidelines-passed over guidelines-tested. This makes brand and accessibility validation *native to the generation platform* rather than a bolt-on proofing step - arguably the most structurally important recent product move in the whole category, because it collapses "generate" and "check" into one system rather than two.

The qualifier that matters: accessibility scope is ADA Level AA *copy/text only* - four of five checks are text properties (contrast, jargon, abbreviations, readability level), and alt text is checked for *existence*, not accuracy. This is not equivalent to axe/WAVE-class WCAG tooling and should not be sold internally as such. The compliance score is a deterministic pass/fail ratio, not an AI confidence score - a related claim asserting otherwise was tested and rejected.

USE CASE 6 - UNCLOSED BY EVERY VENDOR EXAMINED

Your journey-validation case - an agent that clicks a URL or QR code, renders the destination, screenshots it, and compares against the legally approved version - is **not closed by any shipped product found in this research.** Filestage's Barcode & QR Agent checks reachability and machine-readability only; it never renders the landing page or compares it against an approved artifact. Ziflow has both halves of the primitive - it renders live websites at any resolution and does pixel-level version comparison - but applies them to assets a human uploads into a proof, not to destinations an agent discovers by traversing a code. Its brand-standards module (which would catch logo/font/color drift on a landing page) is still "coming soon." This is the single clearest, most actionable white space in the entire report.

08 Segment E - The system-of-record layer

Where creative assets actually live, and what real integration surfaces exist for pushing them into review and pulling annotations back out.

THIS SECTION IS AN HONEST GAP, NOT A FINDING

Across three research passes, the dedicated agents assigned to AEM Assets/Sites APIs, Adobe Workfront and Workfront Proof's standalone status, the Aprimo REST API, and real Jira integration patterns for creative QA either returned nothing or failed mid-task on a session-limit error before reporting back. **This report cannot tell you which proofing vendors publish native AEM, Aprimo, Workfront, or Jira connectors** - which is directly your use case #9.

The one data point that did surface: GlobalVision's Verify API integrates natively with Hybrid CLOUDFLOW and Esko Automation Engine - both packaging-prepress systems, notably *not* any of the four systems you named. If that pattern holds generally, it would mean the deep-comparison vendors (Segment B) live in a different integration world than your marketing system of record entirely - but that is a hypothesis from one data point, not a verified finding. This is worth a direct question to Ziflow, Filestage, and Aprimo sales engineering rather than further open research; vendor integration marketplace pages are the fastest path to a real answer.

What we do know, generally

INFERENCE Standard practice in this segment is for a proofing tool to be triggered by an asset landing in the DAM/CMS (via webhook or manual push), route it through review, and write approval status back as metadata - but we do not have a verified, vendor-specific description of this for AEM, Aprimo, or Workfront specifically. Treat any vendor's marketing-page claim of a "native AEM integration" as unverified until you've seen the actual connector documentation, given how often "integration" on a marketing page turned out to mean a Zapier-style webhook rather than a first-class connector elsewhere in this research.

09 Segment F - Financial-services marketing compliance

The regulatory equivalent of pharma MLR review for banks, broker-dealers, RIAs, and insurers. This section is the best-sourced regulatory material in the whole report - every claim below comes from FINRA or SEC primary text, not a law firm summary.

FINRA Rule 2210 - the regulation that creates the market

VERIFIED, FINRA.org Rule 2210 sorts every firm communication into three categories with a bright-line test: a **retail communication** is anything distributed to more than 25 retail investors in any 30-day period; **correspondence** is the same to 25 or fewer; **institutional communication** goes only to institutional investors. This threshold is the practical trigger for which review regime applies.

CATEGORY	PRE-USE APPROVAL?	GOVERNING REQUIREMENT
Retail communication	**Yes** - one registered principal, before use or filing	Rule 2210(b)(1)(A)
Correspondence	No - supervised after the fact	Rules 3110(b), 3110.06–.09
Online interactive forum posts	No - supervised like correspondence	Rule 2210(b)(1)(D)
Institutional communication	No individual pre-approval; written procedures required	Rule 2210(b)(3)

VERIFIED Filing with FINRA's Advertising Regulation Department runs on 10-business-day windows: *pre-filing* at least 10 days before first use for new members' first-year public-media communications, non-generally-published fund performance rankings, and security-futures communications; *post-filing* within 10 days of first use for retail communications concerning registered investment companies, public DPPs, CMOs, and structured/derived products.

VERIFIED Recordkeeping ties directly to SEC Rule 17a-4(b): firms must retain the communication itself, dates of first and last use, **the name of the approving principal and the date of approval**, source data for any statistics, and any FINRA review letters. Critically, this obligation attaches to *content*, not channel - it follows business communications onto text messages, chat apps, and personal devices, and firms must have capture capability *before* permitting use of a channel at all. That single requirement is the actual compliance driver behind archiving/surveillance vendors (Smarsh, Global Relay, Proofpoint) as a category distinct from pre-publication review tools - they exist to satisfy this clause, not the approval clause.

VERIFIED Content standards (Rule 2210(d)(1)): fair dealing and good faith, fair and balanced, a sound basis for evaluating the facts; no false, exaggerated, unwarranted, promissory, or misleading claims; no predictions or projections of performance (three narrow enumerated exceptions).

VERIFIED FINRA Regulatory Notice 24-09 (June 27, 2024) states FINRA rules are technology-neutral: the existing content standards apply in full to AI-generated and LLM-drafted communications, and a firm cannot outsource compliance responsibility to a third-party AI vendor. FINRA created no new AI-specific rule - it simply confirmed the old ones already cover this.

SEC Marketing Rule (206(4)-1) - the adviser-side equivalent

VERIFIED Effective May 4, 2021, compliance date November 4, 2022. Core content standard: any advertisement showing gross performance must show net performance with equal prominence, over the same period, using the same methodology. A March 2025 FAQ update loosened this slightly, permitting "extracted performance" of a portfolio subset without a matching net figure under specific conditions - a real 2025 easing worth knowing about if you're building rules against the 2022 baseline.

VERIFIED Unlike FINRA, the Marketing Rule imposes **no pre-use principal approval requirement at all** - the SEC proposed one and dropped it in the final 2020 rule. This corrects a common assumption; if a vendor implies federal law mandates pre-approval for RIA marketing, that's not accurate.

Enforcement - what actually gets fined, and for what

VERIFIED, FINRA AWCs The dominant enforcement pattern in 2024–2025 is **influencer marketing**: Robinhood \$26M + \$3.76M restitution (misleading, exaggerated, promissory influencer posts, undisclosed paid placements, ~75 influencers, \$2.7M paid); M1 Finance \$850K; Webull \$1.6M; Moomoo \$750K; Open to the Public \$350K; TradeZero \$250K; Cobra Trading \$200K. FINRA's own crypto-communications sweep found potential Rule 2210 violations in **roughly 70% of 500+ crypto retail communications reviewed.** Firms own what influencers post about them - there is no "we didn't write it" defense.

VERIFIED, SEC orders Three sweeps since 2023: hypothetical performance without adequate policies (\$850K combined, 9 advisers, Sept 2023 - two of the nine were separately charged just for failing to retain copies of their own ads); unsubstantiated capability/AUM claims (\$200K combined, 5 advisers, April 2024); and stale or undisclosed testimonials and third-party ratings (\$1.24M combined, 9 advisers, Sept 2024 - four firms cited ratings more than five years old without disclosing the date, which is exactly the class of defect an automated pre-publication check could catch mechanically).

10 How enterprise review actually works

The operational reality behind every vendor pitch in this report. Two honest findings up front: there is no independent data on cycle times or rework rates anywhere - every statistic in circulation traces back to a company that sells proofing software - and the regulatory material, by contrast, is fully verifiable and does most of the real explanatory work.

Roles, in order

```
brief owner (brand/marketing mgr)
  ↓
creative ops / traffic — opens job, assigns resources
  ↓
creative director → art director + copywriter — concept
  ↓
internal creative review  (never counted in cycle-time stats)
  ↓
studio / production artist — mechanical: live text, bleed, fonts, color, versioning
  ↓
marketing ops — routes into review system
  ↓
brand review  →  legal counsel  →  regulatory/compliance  →  (medical, if pharma)
  ↓
final routing for signature
  ↓
release to channel  →  print/production vendors preflight
```

A correction worth carrying into any internal pitch

VERIFIED There is **no FDA regulation** defining a pharma MLR/PRC committee, its membership, or its procedures - FDA regulates the output (the piece and its 2253 filing), not the process that produced it. The multi-week pharma review cycle is self-imposed risk management, not a regulatory mandate. Similarly, FINRA requires only *one* principal, pre-use, no committee. The SEC Marketing Rule requires no pre-use approval at all. If your internal narrative for why review is slow leans on "the regulation requires it," check which regulation actually says that - often none does, and the multi-reviewer committee is an enterprise's own risk-aversion, which means it's also the enterprise's own lever to pull.

How many rounds - and why every number is vendor-sponsored

STATISTIC	SOURCE	FLAG
Avg. 8 days, 3+ versions to sign-off	Filestage, 2023 (n=366, self-selected)	VENDOR
57% need 3–5 versions; 25% need 6+	Ziflow/AMA, 2023 (no n disclosed)	VENDOR
Pharma MLR: 1.5 avg. circulations (2025), **up** from 1.7 (2021)	Vodori Pepper Flow benchmarks	VENDOR - but see below
Recirculation cause: typos 23.8% (largest single cause), claims 13.5%, branding 9.9%	Vodori, 2025	VENDOR

INFERENCE One data point here is worth taking seriously precisely *because* it's vendor-sponsored and runs against the vendor's own sales narrative: Vodori's own telemetry shows pharma MLR job duration getting *worse*, not better - 13.6 to 13.8 to 14.8 days across 2021, 2024, and 2025, with review duration climbing from 6.6 to 7.6 days. And the single largest driver of extra rounds is typos, not claim disputes - nearly a quarter of expensive regulatory recirculations are, at bottom, proofreading failures.

The annotated PDF as artifact of record - why it survives

VERIFIED Three separate regulatory mechanics converge on the annotated, signed PDF as the legal record, and none of them is "because enterprises are old-fashioned":

- **21 CFR 11.10(b)** requires records be producible in human-readable form for inspection.
- **21 CFR 11.50** requires the *meaning* of a signature (review, approval, authorship) be embedded in that human-readable rendering itself - not just logged separately - and 11.70 requires it can't be excised or transferred to falsify the record.
- **17 CFR 240.17a-4(b)(4)** (the financial-services equivalent) requires firms preserve "communications sent... *and any approvals thereof*" - the approval is itself a record, not metadata about one.

INFERENCE The annotated PDF survives because it's the only object that fuses three things regulators separately demand: what the reviewer saw (the rendered page), what they said (the annotation, positioned on the thing it refers to), and who said it and when. A comment thread has the second and third but not the first; an email approval has the third but neither of the others.

> **THE STRUCTURAL PROBLEM NOBODY MENTIONS**
>
> VERIFIED, prepressure.com By PDF/X production standard, annotations must sit outside the printable/bleed area - a "small icon of a post-it note in the middle of a page on a printing plate" is a real, documented failure mode. This means the review-chain PDF (with annotations) and the production PDF/X file **cannot be the same file**. Every enterprise runs two parallel PDF lineages, reconciled by hand, unlogged. Enfocus's "Certified PDF," which tried to embed a self-auditing change log directly into the file, never made it into the PDF spec and industry support has faded - the one serious attempt to fix this died.

Practitioner sentiment - ranked failure modes

1. **Comments don't carry forward between versions.** Universal complaint across Ziflow, Frame.io, Veeva. Reviewers re-annotate from scratch each round unless an admin-gated feature most agencies don't know exists is turned on.
2. **Can't ingest an already-marked-up PDF as the initial version** (PageProof), forcing a choice between the annotated-PDF chain and the tool's own chain.
3. **Annotations obscure the underlying work** during review.
4. **Reviewers who won't use the tool.** The winning fix in one 112-comment agency thread was literally "use a Google Doc."
5. **Per-reviewer licensing pushes teams into shared logins**, which destroys the named-authenticated-user property every audit-trail regulation depends on. A pricing decision is silently a compliance decision.
6. **Notifications are simultaneously broken and overwhelming** - the same reviewer, the same review, listing "notifications" as both a pro and a con.
7. **Mobile review is universally unusable** across every vendor checked.
8. **Compare mode fails at exactly the job it exists for** - add or remove a page and page-by-page comparison breaks.
9. **Nobody owns proofreading.** Typos are the #1 driver of MLR recirculation, and proofreader budgets are the first thing cut.
10. "**Consolidated markup" has no standardized term or artifact in marketing.** The rigorous version - a Comment Resolution Sheet with a permanent ID, response codes (Accepted / Rejected / Clarification Needed), and closure only by the original reviewer - is a real, defined practice in architecture and construction. Marketing has the problem and not the artifact.

11 The cost of getting it wrong

Every incident below is drawn from a regulator's own press release, order, AWC, final notice, or ruling - not a law-firm blog. This is the evidence base for "why does anyone pay for proofing software."

WHO	WHEN	AMOUNT	WHAT THE CREATIVE/DISCLOSURE ACTUALLY SAID OR OMITTED
Amazon (Prime)	Sep 2025	$2.5B	Enrollment flow didn't clearly disclose recurring-subscription terms at the point of purchase
Publishers Clearing House	Jun 2023	$18.5M	Clarifying copy in "small, light font, below the call-to-action buttons"
Robinhood (FINRA)	Mar 2025	$26M + $3.76M restitution	Influencer posts: misleading, exaggerated, promissory claims; undisclosed paid placements
Williams-Sonoma	Apr 2024	$3.175M	"Crafted in America" label on China-made mattress pads
Kubota	Jan 2024	$2M	Labels never updated after manufacturing moved overseas
Apex Clearing / SoFi (FINRA)	2023/2025	$3.2M / $500K	Disclosure document distributed to 5M+ investors stated a loan fee they did not actually receive
Credit Suisse (FCA, UK)	2014	£2.398M	Shrank the minimum-return font to match the maximum-return font in a brochure
Floris Huisamen, LCF compliance director (FCA)	Feb 2024	£31,800 + ban	Personally signed off promotions FCA called "an ineffective tick-box exercise"
SEC testimonial sweep (9 advisers)	Sep 2024	$1.24M combined	Third-party ratings cited from 2001–2004 without disclosing the date
University of Phoenix	Dec 2019	$191M	Ads implied corporate partnerships (Microsoft, AT&T, Twitter logos) that did not exist

Five patterns worth building an internal case around

1. **The disclosure often existed and still failed - on layout, not wording.** Floki Inu's risk warning was present but too small relative to the headline. Nationwide's qualification was on screen but "likely to be missed." Credit Suisse's fine cites the specific act of shrinking a font. Regulators adjudicate visual hierarchy, not just text.
2. "**Nobody re-read it" is a recurring, expensive failure.** Kubota's $2M was labels never updated after production moved offshore. Santander (FCA, £12.4M, partial) ran unchanged brochures for eight years. Stale statistics and expired ratings recur across nearly every sweep.
3. **The sign-off process is itself an enforcement target** - not just the content. LCF's compliance director was personally fined and banned for how he approved, not what was approved.
4. **Third-party creative is your creative.** Seven-figure FINRA fines landed on firms for what influencers posted, not what the firms themselves wrote.
5. **The base rate is brutal.** FINRA found potential violations in ~70% of 500+ crypto retail communications reviewed. The FCA saw 19,766 promotions amended or withdrawn in 2024 - a 97.5% increase year over year. This is not a tail risk; it's the modal outcome when marketing material meets a regulator.

12 The AI question - real vs. vaporware

The most important finding in this entire research effort, stated once, plainly: **agentic QA in this category today is agentic detection with human adjudication, not agentic sign-off.** Every single GA product examined stops short of autonomous approval.

What failed verification

EVERY QUANTIFIED VENDOR ROI CLAIM TESTED WAS KILLED ON ADVERSARIAL REVIEW

- Revisto's "$15M+ saved per brand annually" - failed 0-3
- AdClear's "88% reduction in marketing review time" - failed 0-3
- Puntt's "six months to a few weeks" review-cycle compression - failed 0-3
- Veeva's "eliminate 70%+ of manual MLR labor within five years" - survived *only* as an attributed, safe-harbored, forward-looking projection, with testimonials sourced to "POC users"

INFERENCE No pilot data, no customer benchmark, no independent validation exists for any of these claims - and notably, no analyst pushback either. That's absence of rebuttal, not confirmation. There is no independent benchmark anywhere in this category for AI review accuracy, false-positive rate, or time saved. A buyer today genuinely cannot compare vendors on evidence.

What's real

- **Down-market distribution.** Rules-based compliance checking that used to require a $195M-scale specialist vendor (GlobalVision) now ships as metered credits inside a $329/month proofing tool (Filestage, Ziflow). The capability didn't get smarter so much as it got cheap and common.
- **Structural decomposition of the rule problem.** Messagepoint's Black Text / Blue Text split is a genuine advance in how to *express* a validation rule, independent of whether an LLM or a deterministic engine executes it.
- **The deliberate rejection of ML in the highest-stakes tier.** Argos's visual-regression engine explicitly chooses deterministic pixel diffing over ML "for auditability" - a direct counter-signal to "AI review is obviously the better answer." In regulated review specifically, a tool that says "here are the differences, you decide" is defensible; a tool that says "this is compliant" inherits the liability for being wrong.

THE DESIGN PATTERN EVERY SERIOUS VENDOR HAS CONVERGED ON

Schlafender Hase, twenty years ago: "*TVT shows you all differences. The expert determines what is right or wrong.*" Ziflow, 2026: outputs "AI-suggested" comments. GlobalVision, 2026: outputs a risk score and flags. Filestage, 2026: agents output suggestions for a human reviewer. Adobe GenStudio: pass/fail per guideline, reviewed by a human before publish. Nobody who takes liability seriously is shipping autonomous sign-off, and that consistency - across vendors who compete with each other and have every incentive to differentiate - is itself evidence that it's the right design, not an accident of the market being early.

13 Mapping your use cases

Your nine specific jobs, matched against what exists today.

YOUR USE CASE	CLOSEST EXISTING ANSWER	STATUS
Side-by-side proofing: email, direct mail, post-login, PDF, image, Excel, HTML	Ziflow ingests all these formats on its free tier with pixel-level compare	SOLVED at ingest
Image AND text comparison in one pass	GlobalVision / Schlafender Hase do this natively for print & pharma packaging	PARTIAL
Validate against a rules checklist (brand + legal)	Ziflow ReviewAI, Filestage Review Agents, Adobe GenStudio Content Check	MOST MATURE
Validate channel-coded asset vs. annotated legal PDF + copy deck	Schlafender Hase TVT (cross-format text compare); Messagepoint MARCIE (Black/Blue Text)	PARTIAL, no direct product
Validate many versions/variants for different comms	Messagepoint MARCIE Assure's Black/Blue Text model - domain-locked to Medicare	PATTERN EXISTS, NOT GENERAL
Journey validation: click QR/URL → render → screenshot → compare vs. approved	Nobody. Primitives exist separately (Ziflow renders + diffs; Filestage checks QR reachability)	UNCLOSED
Validate dynamic variables vs. Excel data matrix	Messagepoint validates plan variables against PBP data - the direct analogue, Medicare-only	PATTERN EXISTS, NOT GENERAL
No-code admin console for rules/output/report config	Not verified anywhere as a standalone capability	UNCLOSED / UNRESEARCHED
Integrations with AEM, Aprimo, Jira	Unresearched this pass - GlobalVision's only known integrations are prepress systems, not these	GAP - ask vendors directly

14 White space

What is genuinely unbuilt, stated with the evidence behind each claim.

1. **No independent benchmark for AI review accuracy exists.** Every vendor efficiency claim tested failed adversarial verification. A market-defining opportunity: be the neutral evaluator, or be the vendor willing to publish a real number.
2. **Journey/destination validation is unclosed by everyone.** The rendering and pixel-diff primitives exist separately at Ziflow; nobody has wired traverse-a-code → render-destination → compare-to-approved into one agent.
3. **Visual-regression-grade tooling has not been pointed at marketing creative.** Applitools, Percy, Chromatic, and Argos-class technology is mature and battle-tested in software QA. Nobody found in this research applies it to marketing assets across channels.
4. **The comment-resolution artifact marketing lacks.** Construction and engineering have a rigorous, closeable-only-by-reviewer Comment Resolution Sheet with permanent IDs and response codes. Marketing has the identical underlying problem and no equivalent artifact - a borrowable pattern, not a novel invention.
5. **General-purpose variant validation.** Messagepoint solved this for one document type (Medicare ANOC/EOC). The Black/Blue Text decomposition generalizes; nobody has generalized the product.
6. **No-code rule/output configurability**, as a distinct, verifiable capability, was not found described anywhere - every vendor examined ships fixed check types (grammar, forbidden terms, barcode) rather than a configurable rule engine an admin can extend without code.

15 Business model & the money

The universal pattern in Segment A

VERIFIED Base proofing - proofs, reviewers, storage, versions - is sold as unlimited across every vendor examined (Ziflow, PageProof, Filestage). AI/compliance is *always* the metered or gated monetization axis: PageProof steps 250/500/1,000 PI credits by tier while proofs stay unlimited on the same page; Filestage meters Review Agents at 50 non-rollover credits/month, gated off Free and Starter entirely; Ziflow gates ReviewAI to Enterprise custom pricing only; GoVisually sells its compliance suite as a flat $499/month superset tier, a ~5x step off its base Pro plan. The commodity surface itself is not defensible - the AI layer is where every vendor is trying to capture margin.

Segment B and F: opaque, sales-led, enterprise-only

GlobalVision, Schlafender Hase, Messagepoint, and every financial-services compliance vendor examined publish **no public pricing whatsoever**. Given GlobalVision alone sold for ~$195M, the per-seat or per-enterprise economics in this tier are presumably substantial, but no figure is publicly available.

Market size - the honest gap

> **NO CREDIBLE ANALYST ESTIMATE EXISTS FOR THIS CATEGORY, IN ANY OF THREE ATTEMPTS**
>
> Gartner, Forrester, and IDC did not yield a market-size figure for online proofing, creative operations, marketing resource management, digital asset management, customer communications management, or marketing-compliance software specifically, across three separate research passes. Any number you see attributed to Grand View Research, MarketsandMarkets, or similar publishers should be treated as a research-mill estimate of unverified methodology, not a citable market size. We are stating this as a real absence in public data, not a search failure on our part - this may simply not be a category any credible analyst has sized as a standalone segment yet.

What consolidated in 2024–2026

The dominant 2024–2026 event in this specific segment (marketing creative proofing, as distinct from the pharma/finserv-compliance landscape covered separately) is **consolidation, not new-entrant funding**: Veralto/GlobalVision (~$195M, April 2026), Validity/Litmus (April 2025), and Sinch retiring Email on Acid into Mailgun Inspect (mandatory migration from June 2026). Incumbents are absorbing point solutions rather than being disrupted by AI-native challengers.

16 Glossary

Including, honestly, the terms this research could not resolve. Where a definition is INFERENCE rather than a sourced fact, it's marked as such - do not treat it as authoritative without checking with whoever uses the term inside your own organization.

TWO TERMS FROM YOUR OWN USE-CASE LIST WE COULD NOT SOURCE

CAF UNRESOLVED - the dedicated research agent for this term failed on a session-limit error before returning results. Plausible expansions, none verified: Creative Approval Form, Content Approval Form, Campaign Approval Form. If this is house vocabulary from a specific client or employer, it may not be documented publicly at all.

MC UNRESOLVED - same failure. Plausible expansions: marketing communication, member communication, material code. The right reading changes what your use case actually means - worth confirming directly rather than guessing.

Copy deck INFERENCE
The document containing the approved words for a piece of creative - typically written by a copywriter, held in Word/Google Docs/Excel, and used as the reference during proofing. A dedicated research pass on this term failed to complete; treat this definition as working shorthand, not a sourced industry standard.

Channel code / channel coding INFERENCE
Embedding tracking, offer, or campaign identifiers into a produced asset so response can be attributed back to the specific version and channel. Related to source codes, key codes, and offer codes in direct marketing. Not independently sourced this session.

Version / variant matrix INFERENCE
A grid tracking which combination of segment, offer, channel, and geography each produced variant of a campaign corresponds to. Not independently sourced this session.

Annotated PDF
A PDF with reviewer comments/markup layered on top of the content, functioning as the legal artifact of record in regulated review - see §10.

Comment resolution log
A structured record of every review comment, its response, and its closure status. A defined, rigorous practice in architecture/construction (the "Comment Resolution Sheet"); not standardized in marketing.

Deterministic vs. ML-based comparison
Deterministic: fixed-algorithm pixel or character comparison with explainable, auditable output (GlobalVision, Argos). ML-based: a trained model judges whether a difference matters (Applitools' "Visual AI" positioning; largely unverified in this research).

False positive
A flagged difference that isn't actually a defect - anti-aliasing artifacts, font-rendering noise, or dynamic content are the most commonly cited causes in visual regression testing.

Retail communication / correspondence VERIFIED
FINRA Rule 2210 terms: distributed to more than 25 retail investors in 30 days (retail communication, requires pre-use principal approval) versus 25 or fewer (correspondence, supervised after the fact).

Principal approval VERIFIED
FINRA's pre-publication sign-off requirement - one registered principal, before use. Not a committee requirement; that's an enterprise's own risk-management addition.

PDF/X
A print-production PDF standard that prohibits annotations inside the printable/bleed area - the reason review-chain PDFs and production PDFs cannot be the same file.

Black Text / Blue Text
Messagepoint MARCIE Assure's rule taxonomy: text that must match verbatim (Black) versus text that must be present only if a variant's specific attributes require it (Blue).

17 Research ledger

What this report is confident about, what it is honestly unsure of, and what's worth your own direct follow-up rather than more open research.

Solid - built on primary sources

- FINRA Rule 2210 and SEC Marketing Rule mechanics (§09) - sourced to FINRA.org and SEC orders directly, zero vendor-copy dependency.
- The Veralto/GlobalVision acquisition (§05) - sourced to an SEC 10-Q filing.
- The 40+ enforcement incidents in §11 - every one sourced to the regulator's own release, order, or ruling.
- Segment A pricing and AI-gating patterns (§04) - verified against raw vendor HTML, not marketing summaries.
- Visual regression mechanism details (§07) - verified against open-source documentation and shipping code (package.json dependencies), not just marketing pages.

Genuinely open - not a search failure, an absence in public data

- Market size for the combined category (§15) - three attempts, zero credible analyst figures.
- AEM/Aprimo/Workfront/Jira integration surfaces (§08) - two dedicated research attempts failed before returning results.
- AI brand governance vendors (Writer.com, Grammarly, Acrolinx, Frontify) - zero verified claims across all passes.
- The FinServ compliance *vendor* landscape specifically (Saifr, Red Oak, PerformLine, Smarsh, Hearsay) - the regulation is well-sourced (§09); which vendor does pre-publication review versus post-hoc archiving is not.
- CAF and MC (§16) - unresolved; worth five minutes with whoever wrote your use-case list rather than more research.

IF YOU WANT THIS FILLED IN

The fastest path on the integration-surface gap and the FinServ vendor landscape is direct vendor conversations (Ziflow, Filestage, Aprimo sales engineering) rather than more open research - those are the kind of specifics that live on gated partner pages and sales decks, not public documentation. The market-size gap may simply not be answerable; treat any number you're handed with real skepticism.